MONASH University

The Confidence Machine

Week 6 — Probability & Bayes' Rule

Julian Garcia

FIT1061 — Introduction to Artificial Intelligence

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MILESTONE

Watson

IBM · 14 February 2011



Ken Jennings, Watson, and Brad Rutter on the *Jeopardy!* stage, 2011.

▶ Watch: Ken Jennings concedes to Watson

A quiz show — and why it is hard for a machine

The game

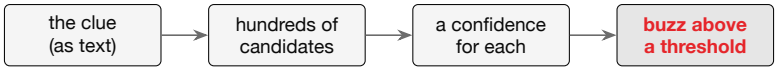
- You are given the **answer**; you must reply with the **question**.
- Clues are full of puns, wordplay, and ambiguity.
- Any topic at all — open-domain trivia.
- Fastest correct buzz wins.

Why that is hard for a machine

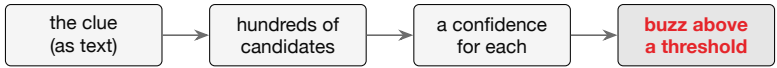
- Read the clue.
- Guess what **kind** of thing the answer is.
- Search millions of documents.
- Beat expert humans on the buzzer.

Questions on any topic, at speed — **very hard for a machine**.

What Watson actually did

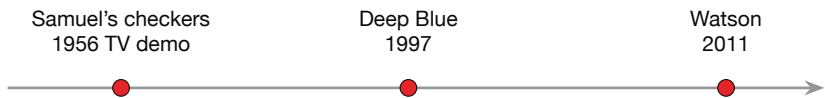


What Watson actually did



Watson was never certain. It was **probably right**.

Impressive — and it made mistakes no human would



An ordered sequence of milestones — not drawn to scale.

Answered **"Toronto"** in the category "U.S. Cities" — a **Canadian** city.

Repeated **"1920s"** just after another player gave that answer and was told it was **wrong** — it could not hear them.

How does a machine act on beliefs
it **is not certain** about?

The history

1763	Update a belief when evidence arrives (Bayes' rule, after his death)
2002	The same idea sorts your email ("A Plan for Spam")
2011	The same idea wins a quiz show (Watson — hundreds of clues)
2022	The same idea predicts your next word (today's chatbots)

Probability — the **language of uncertainty** — from Bayes to today's chatbots.

Poll

A disease affects **1%** of people. A test is **99% accurate**. You test **positive**.

("99% accurate" = it is right 99% of the time on sick and on healthy people alike.)

What is the chance you actually have the disease?

99% **90%** **50%** **1%**

Vote now — scan the code for your campus, *or* type the address.

Melbourne
pollev.com/juliangarcia

Malaysia
pollev.com/onghueyfang767

Most people say about **99%**.

Refresher — the one move from your prep

10 days. It rained on **3**. Of the 10 days, **4** were cloudy, and **3** of those cloudy days rained.

	rain	**no rain**
cloudy	3	1
not cloudy	0	6

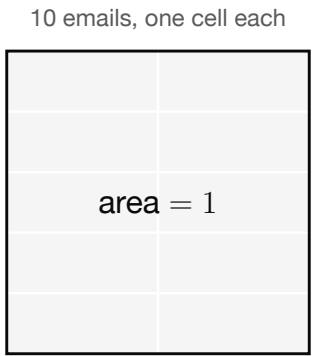
What is $P(\text{rain})$? And $P(\text{rain} \mid \text{cloudy})$?

Probability is area



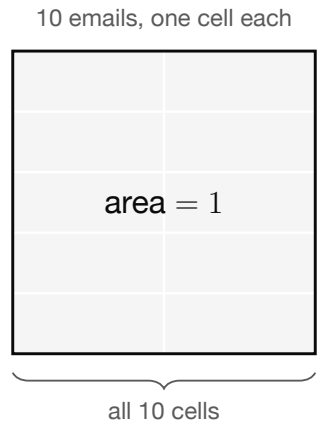
A region's area is its probability. **Area is made of cells**: 1 cell $=$ 1 email.

Probability is area



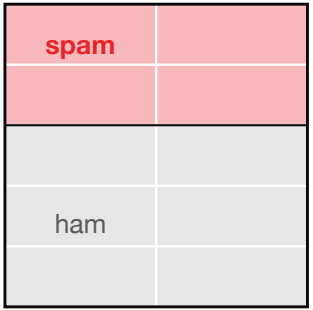
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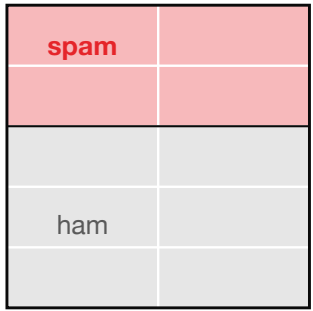
A region's area is its probability. **Area is made of cells**: 1 cell = 1 email.

Spam is a region



$P(\text{spam})$ = the spam share of all 10 cells $= 4/10 =$ **0.4**.

Spam is a region

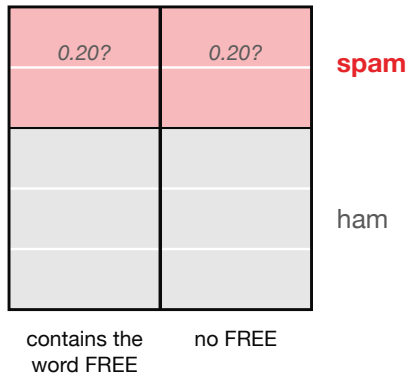


$P(\text{spam}) = \mathbf{0.4}$

4 of the 10 cells

$P(\text{spam})$ = the spam share of all 10 cells $= 4/10 = \mathbf{0.4}$.

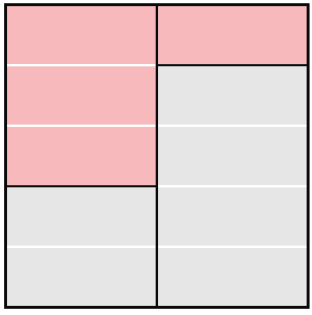
Add the FREE cut



The red band is **unchanged**. We only add the FREE cut.

If the word FREE were not associated with spam, each half would be $0.5 \times 0.4 = 0.20$ (2 of the 10 cells).

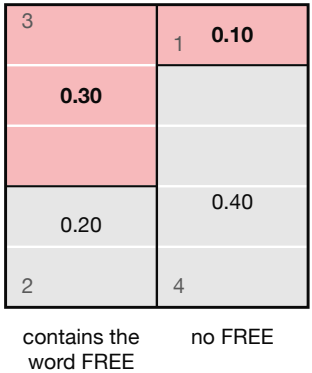
Spam clusters in FREE



Spam concentrates in FREE, so the overlap is **0.30**, above the even-split guess of 0.20.

same red, re-sorted: $0.30 + 0.10 = 0.40$ (that is 3 cells + 1 cell).

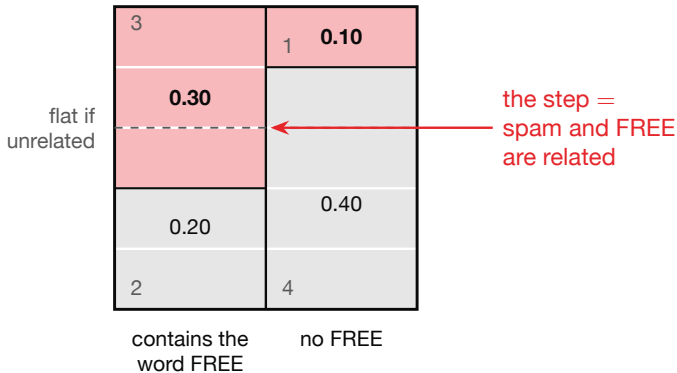
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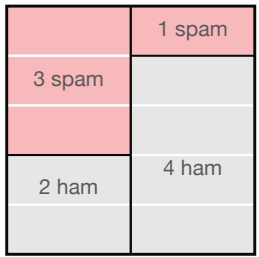
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A new email says FREE



A new email **contains the word FREE**.

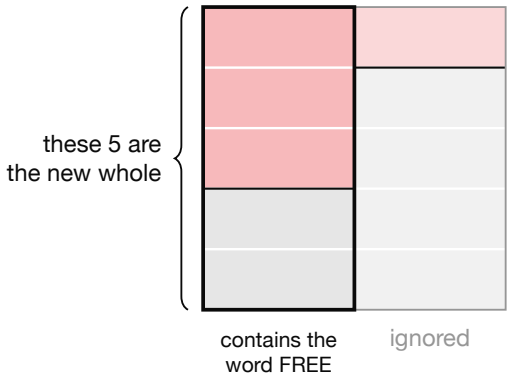
Any email is spam with probability **0.4** (4 of 10).

The thing we want is the revised chance:

$$P(\text{spam} \mid \text{FREE})$$

The bar "|" means **given**: the chance it is spam, given it contains FREE.

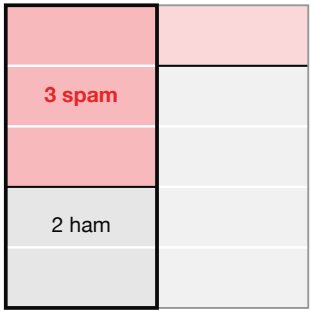
Look only at the FREE emails



FREE tells us something, so **look only at** the 5 FREE emails.

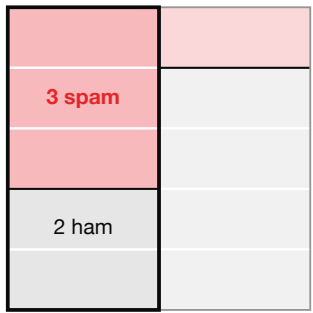
The rest are dimmed, not gone. These 5 become the new “whole”.

Of those 5, how many are spam?



Knowing FREE increases the chance of spam from $4/10 =$ **0.4** to $3/5 =$ **0.6**.

Of those 5, how many are spam?



5 FREE emails

3 of the 5 are spam:

$$\frac{3}{5} = \mathbf{0.6}$$

Knowing FREE increases the chance of spam from $4/10 = \mathbf{0.4}$ to $3/5 = \mathbf{0.6}$.

You already do this

It is **cloudy** this morning. Will it rain today?

You do not think about every day of the year.

You **look only at cloudy days** and ask: how many of those brought rain?

The same two steps: look only at the cases that match, then measure.
Now back to the emails.

This is conditional probability

We read $P(\text{spam} \mid \text{FREE})$ straight off the areas:

$$P(\text{spam} \mid \text{FREE}) = \frac{\text{spam and FREE}}{\text{all FREE}} = \frac{3}{5} = \frac{0.30}{0.50} = \mathbf{0.6}$$

This is conditional probability

We read $P(\text{spam} \mid \text{FREE})$ straight off the areas:

$$P(\text{spam} \mid \text{FREE}) = \frac{\text{spam and FREE}}{\text{all FREE}} = \frac{3}{5} = \frac{0.30}{0.50} = \mathbf{0.6}$$

Conditional probability

$$P(A \mid B) = \frac{\text{area}(A \cap B)}{\text{area}(B)}$$

Look only at B (the rest is dimmed, not gone), then measure A's share.

Poll

We started believing a message was **40%** likely to be spam. Then we saw it contains the word **FREE**.

After that evidence, our belief should:

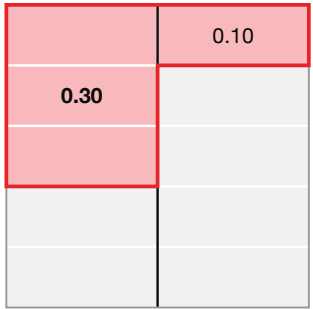
Go up | **Go down** | **Stay the same** | **We cannot say**

Vote now — scan the code for your campus, *or* type the address.

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Malaysia
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Now look only at spam

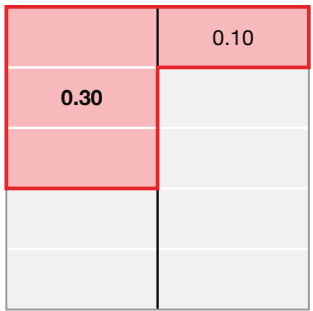


kept: spam only (area 0.40)

$$P(\text{spam} \mid \text{FREE}) = \frac{3}{5} = 0.6 \quad \neq \quad 0.75 = \frac{3}{4} = P(\text{FREE} \mid \text{spam})$$

Two different questions — but multiply each by its own region.

Now look only at spam

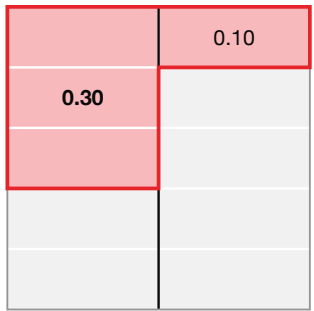


kept: spam only (area 0.40)

FREE's share of the spam region:

$$\frac{3}{4} = \frac{0.30}{0.40} = \mathbf{0.75}$$

Now look only at spam



kept: spam only (area 0.40)

FREE's share of the spam region:

$$\frac{3}{4} = \frac{0.30}{0.40} = \mathbf{0.75}$$

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Two different questions — but multiply each by its own region.

One cell, two readings

The same 0.30 cell, two ways: the FREE area $\times \frac{3}{5}$, or the spam area $\times \frac{3}{4}$.

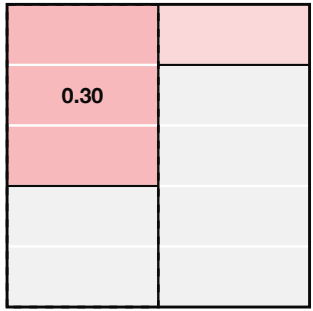


$$P(\text{FREE})\,P(\text{spam} \mid \text{FREE}) \;=\; P(\text{spam})\,P(\text{FREE} \mid \text{spam}) \;=\; \mathbf{0.30}$$

One cell, two readings

The same 0.30 cell, two ways: the FREE area $\times \frac{3}{5}$, or the spam area $\times \frac{3}{4}$.

FREE first (5 cells), then 3 of its 5: $0.5 \times \frac{3}{5} = 0.30$

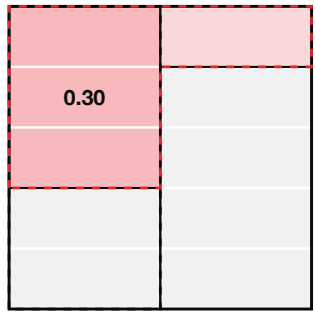


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One cell, two readings

The same 0.30 cell, two ways: the FREE area $\times \frac{3}{5}$, or the spam area $\times \frac{3}{4}$.

FREE first (5 cells), then 3 of its 5:
$0.5 \times \frac{3}{5} = 0.30$



spam first (4 cells), then 3 of its 4:
$0.4 \times \frac{3}{4} = 0.30$

$$P(\text{FREE})\,P(\text{spam} \mid \text{FREE}) \;=\; P(\text{spam})\,P(\text{FREE} \mid \text{spam}) \;=\; \mathbf{0.30}$$

Bayes' rule

One rectangle, two products:

$$P(\text{FREE})\,P(\text{spam} \mid \text{FREE}) \;=\; P(\text{spam})\,P(\text{FREE} \mid \text{spam})$$

Bayes' rule

One rectangle, two products:

$$P(\text{FREE})\,P(\text{spam} \mid \text{FREE}) \;=\; P(\text{spam})\,P(\text{FREE} \mid \text{spam})$$

Divide both sides by $P(\text{FREE})$:

Bayes' rule

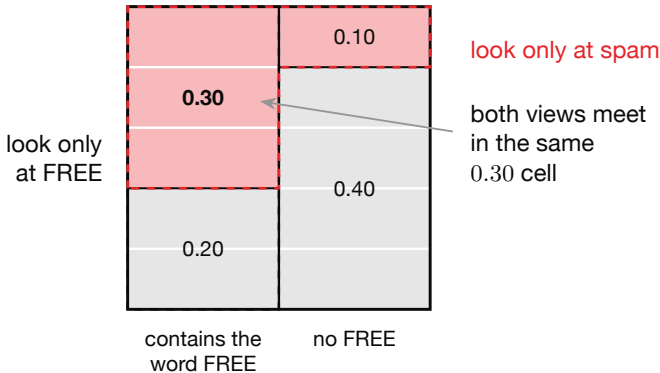
One rectangle, two products:

$$P(\text{FREE})\,P(\text{spam} \mid \text{FREE}) \;=\; P(\text{spam})\,P(\text{FREE} \mid \text{spam})$$

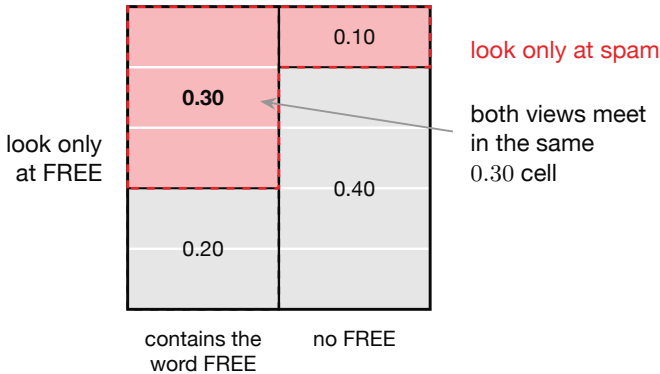
Divide both sides by $P(\text{FREE})$:

$$P(\text{spam} \mid \text{FREE}) = \frac{P(\text{FREE} \mid \text{spam})\,P(\text{spam})}{P(\text{FREE})} = \frac{0.75 \times 0.4}{0.5} = \mathbf{0.6} = \tfrac{3}{5}$$

One area, two measurements



One area, two measurements



Key idea

Bayes' rule is **one area, measured two ways**.

The same 10 emails, as a table

	FREE	no FREE	total
spam	3	1	**4**
ham	2	4	**6**
total	**5**	**5**	**10**

This is the cell grid, rearranged into a table: same 3, 2, 1, 4. You will use the table form in the lab.

The same rule, for any question

You **see** a positive test — the **data**, D.

You **want** to know: are you sick? — the **hypothesis**, H.

The same rule, for any question

You **see** a positive test — the **data**, D.

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You can measure $P(D \mid H)$; you want $P(H \mid D)$.

Bayes flips one into the other.

The same rule, for any question

You **see** a positive test — the **data**, D.

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You can measure $P(D \mid H)$; you want $P(H \mid D)$.

Bayes flips one into the other.

Bayes' rule

$$P(H \mid D) = \frac{P(D \mid H)\, P(H)}{P(D)}$$

$H =$ what you want (sick? spam?); $D =$ what you see (test? FREE?).

Poll — vote again

Same disease, same test: found in **1 in 100** people, test **99% accurate**, and you test **positive**.

What is the real chance you have the disease?

99% **90%** **50%** **1%**

Vote now — scan the code for your campus, *or* type the address.



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The disease puzzle

Spam was **common** (4 in 10). Now try something **rare**.

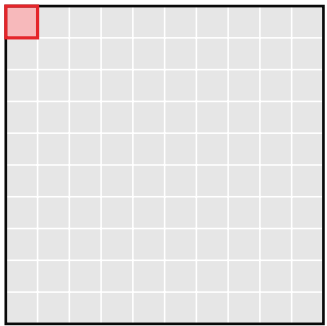


A disease affects **1%** of people. A test is **99% accurate**: 99% of sick people test positive, 99% of healthy people test negative. You test **positive**.

$$P(\text{sick} \mid \text{positive}) = ?$$

We'll **count people** in a grid: **red** = sick, grey = healthy, a dot = a positive test.

10,000 people in one grid



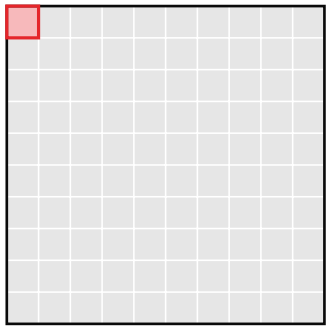
Sick = the 1 red cell
= 100 people = 1%.

Healthy = the other 99 cells
= 9,900 people.

Each cell = 100 people.

Area is made of cells.

10,000 people in one grid



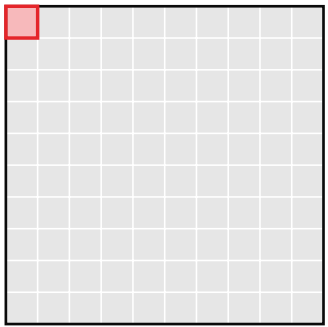
Sick = the 1 red cell
= 100 people = 1%.

Healthy = the other 99 cells
= 9,900 people.

Each cell = 100 people.
Area is made of cells.

Base rate = **1 sick cell in 100** = 1%. Count the red: just one.

The test marks cells



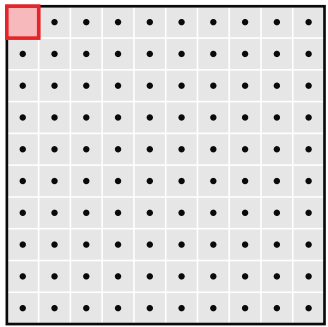
Sick cell: 99 of its 100 test + (the true positives).

Each grey cell: about 1 in 100 tests + (a **false alarm**).

One per grey cell: **99 in all**.

99 true positives, all in one cell. **99** false alarms, one per grey cell.

The test marks cells

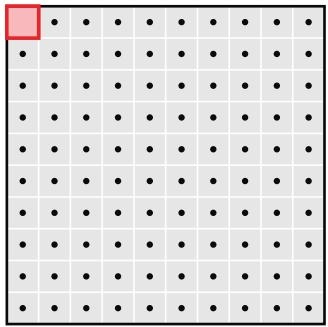


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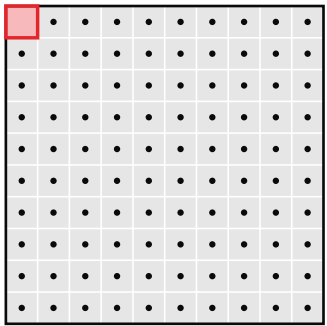
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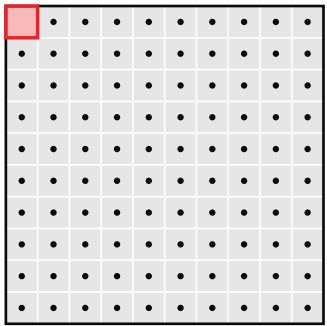
99 true positives, all in one cell. **99** false alarms, one per grey cell.

Count the positives



Count them: how many tested positive? How many of those are really sick?

Count the positives



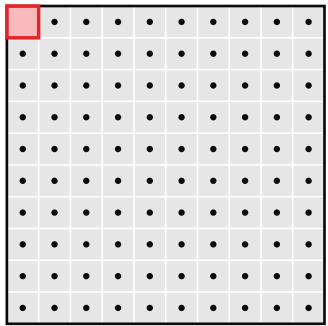
Count them: how many tested positive? How many of those are really sick?

99 real positives
(the one red cell)

99 false positives
(99 grey cells, 1 dot each)

Everyone positive $= 198$.

Count the positives



Count them: how many tested positive? How many of those are really sick?

99 real positives
(the one red cell)

99 false positives
(99 grey cells, 1 dot each)

Everyone positive $= 198$.

$$P(\text{sick} \mid \text{positive}) = \frac{99}{99 + 99} = \mathbf{50\%}$$

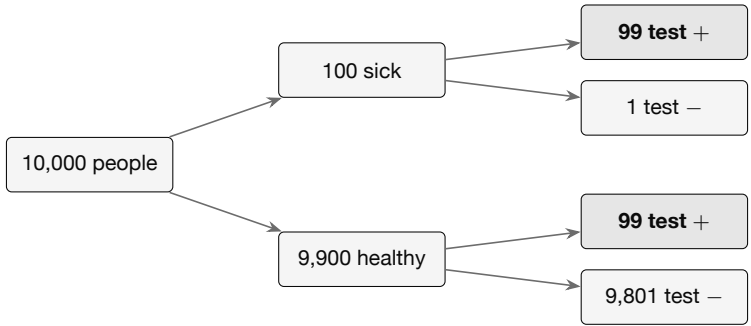
The 99 false alarms match the 99 real ones, so a positive is a **50/50 chance**.

The same count, as a tree



The same two **99**s. You can draw a tree anywhere — no grid needed.

The same count, as a tree



The same two **99**s. You can draw a tree anywhere — no grid needed.

A rare disease can fool a good test

The same 50%, by Bayes' rule. The 99 real over the 198 positives:

$$P(\text{sick} \mid \text{positive}) = \frac{0.01 \times 0.99}{0.01 \times 0.99 \;+\; 0.99 \times 0.01}$$

$$= \frac{0.0099}{0.0198} = \frac{99}{198} = \textbf{50\%}$$

numerator $=$ the 99 real (0.0099); denominator $=$ all 198 positives (0.0198).

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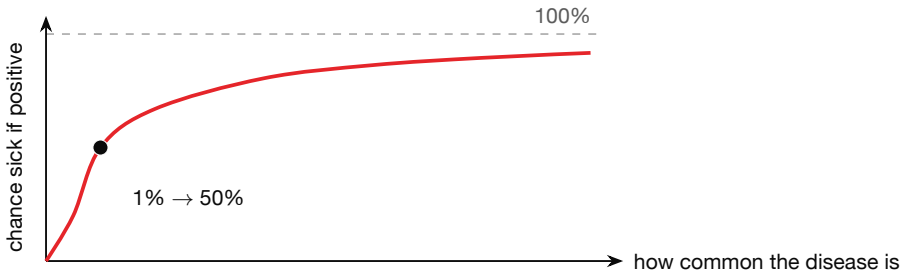
numerator $=$ the 99 real (0.0099); denominator $=$ all 198 positives (0.0198).

Key idea

A **rare** condition can make a positive from even a 99%-accurate test far from certain.

It is 50% *here* because the disease is as rare as the false-alarm rate; rarer, so **still less** trustworthy.

Change how rare the disease is



Axis illustrative — only the 1% → 50% point is exact.

Same 99% test — the answer depends on how **common** the disease is.

The same trap, in familiar places

A rare condition + an accurate test ⇒ most positives are still **false alarms**.

- **Cancer screening.** Most positive mammograms in low-risk groups are false alarms.
- **Security screening.** If the threat is one in a million, most alarms are innocent.
- **Fraud detection.** Fraud is rare, so most flagged purchases are genuine.
- **A "one-in-a-million" DNA match.** In a big city, several innocent people can match.

The last example brings probability into a courtroom — the topic we move to next.

Watson for Oncology

When confidence meets the clinic

The same kind of system that dominated a quiz show was promoted as a way to **revolutionise cancer diagnosis**.

- No independent, published clinical trials.
- High-profile hospital contracts **cancelled**.
- Quietly discontinued after years of over-promising.

Confidence is **not** correctness.

Right **on average** is not the same as right for **this patient**.

Discussion

What goes wrong when we deploy **probabilistic AI** in high-stakes domains?

You will write 150–250 words on this in the lab — use the base-rate trap and Watson for Oncology.

Key takeaways

Four things to keep

1. Probability is **counting**.
2. Bayes flips the conditional — want $P(H \mid D)$, count $P(D \mid H)$.
3. The **base rate** matters — a 99% test can be 50% wrong.
4. Evidence **updates** belief.

In the lab this week

Aiming for a Pass

1. Build the count table from a set of labelled emails.
2. Compute a conditional probability by narrowing the table.
3. Apply Bayes' rule to flip the conditional.
4. Hand-compute the disease-test posterior for your own numbers.
5. Write 150–250 words on deploying probabilistic AI in high-stakes domains.

Aiming further? Think about how you would classify a message with many words at once, not just one.

We can flip the conditional for **one** word.

Real messages have **many**.

We can flip the conditional for **one** word.

Real messages have **many**.

Next: a shortcut that assumes **words act independently**

— so we can classify **many-word** messages.